

A decorative border at the top of the page featuring several stylized, elongated leaves with detailed vein patterns, rendered in a light blue/cyan color against a purple and pink background.

DATA MINING

ASSIGNMENT 4

NAME :- SACHIN

ROLL NO. :- 251256

DEPT :- B.Sc (H.) CS

A decorative border at the bottom of the page featuring several stylized, elongated leaves with detailed vein patterns, rendered in a light blue/cyan color against a purple and pink background.

QUESTION

You are given a transactional dataset from an electric store and a specific list of itemsets (chosen by the user, not generated automatically). Write a program to compute the for each given itemset, and the for given association rules.

```
transactions = [  
    ['Laptop', 'Mouse', 'Laptop Bag'],  
    ['Laptop', 'Keyboard', 'Mouse'],  
    ['Mobile', 'Charger', 'Earphones'],  
    ['Laptop', 'Mouse', 'Keyboard', 'Laptop Bag'],  
    ['Mobile', 'Charger', 'Screen Guard'],  
    ['Laptop', 'Laptop Bag', 'Mouse'],  
    ['Mobile', 'Earphones', 'Screen Guard'],  
    ['Laptop', 'Keyboard', 'Mouse', 'Mouse Pad'],  
    ['Mobile', 'Charger', 'Earphones', 'Screen Guard'],  
    ['Laptop', 'Mouse', 'Mouse Pad']  
]  
  
itemsets = [  
    {'Laptop'},  
    {'Mouse'},  
    {'Mobile'},  
    {'Laptop', 'Mouse'},  
    {'Mobile', 'Charger'},  
    {'Laptop', 'Mouse', 'Keyboard'}  
]  
  
rules = [  
    ({'Laptop'}, {'Mouse'}),  
    ({'Mobile'}, {'Charger'}),  
    ({'Laptop', 'Mouse'}, {'Keyboard'}),  
    ({'Charger'}, {'Mobile'})  
]
```

Q1: Find Support for all itemsets.

Q2: Find support and confidence of all association rules.

Display the data in tabular format.

```

1  from tabulate import tabulate
2
3  transactions = [
4      ['Laptop', 'Mouse', 'Laptop Bag'],
5      ['Laptop', 'Keyboard', 'Mouse'],
6      ['Mobile', 'Charger', 'Earphones'],
7      ['Laptop', 'Mouse', 'Keyboard', 'Laptop Bag'],
8      ['Mobile', 'Charger', 'Screen Guard'],
9      ['Laptop', 'Laptop Bag', 'Mouse'],
10     ['Mobile', 'Earphones', 'Screen Guard'],
11     ['Laptop', 'Keyboard', 'Mouse', 'Mouse Pad'],
12     ['Mobile', 'Charger', 'Earphones', 'Screen Guard'],
13     ['Laptop', 'Mouse', 'Mouse Pad']
14 ]
15
16 itemsets = [
17     {'Laptop'},
18     {'Mouse'},
19     {'Mobile'},
20     {'Laptop', 'Mouse'},
21     {'Mobile', 'Charger'},
22     {'Laptop', 'Mouse', 'Keyboard'}
23 ]
24
25 rules = [
26     ({'Laptop'}, {'Mouse'}),
27     ({'Mobile'}, {'Charger'}),
28     ({'Laptop', 'Mouse'}, {'Keyboard'}),
29     ({'Charger'}, {'Mobile'})
30 ]
31
32
33 def support(itemset):
34     count = sum(itemset.issubset(set(t)) for t in transactions)
35     return count / len(transactions)
36
37
38 # Q1
39 table1 = []
40
41 for itemset in itemsets:
42     table1.append([
43         str(itemset),
44         round(support(itemset), 2)
45     ])
46
47 print("Q1: Support of Itemsets")
48 print(tabulate(table1, headers=["Itemset", "Support"], tablefmt="grid"))
49
50
51 # Q2
52 table2 = []
53
54 for left, right in rules:
55     rule_support = support(left | right)
56     confidence = rule_support / support(left)
57
58     table2.append([
59         str(left) + " -> " + str(right),
60         round(rule_support, 2),
61         round(confidence, 2)
62     ])
63
64 print("\nQ2: Support and Confidence of Rules")
65 print(tabulate(
66     table2,
67     headers=["Rule", "Support", "Confidence"],
68     tablefmt="grid"
69 ))

```

OUTPUT

```
● (.venv) PS D:\DM assignment 1> & "d:\DM assignment 1\.venv\Scripts\python.exe" "d:/DM assignment 1/Practical 4/DM-Asssignment4.py"
```

Q1: Support of Itemsets

Itemset	Support
{'Laptop'}	0.6
{'Mouse'}	0.6
{'Mobile'}	0.4
{'Laptop', 'Mouse'}	0.6
{'Charger', 'Mobile'}	0.3
{'Keyboard', 'Laptop', 'Mouse'}	0.3

Q2: Support and Confidence of Rules

Rule	Support	Confidence
{'Laptop'} -> {'Mouse'}	0.6	1
{'Mobile'} -> {'Charger'}	0.3	0.75
{'Laptop', 'Mouse'} -> {'Keyboard'}	0.3	0.5
{'Charger'} -> {'Mobile'}	0.3	1

```
○ (.venv) PS D:\DM assignment 1> █
```

The image features a decorative background with a watercolor-like texture in shades of purple, blue, and pink. The top and bottom edges are adorned with a row of stylized, dark blue leaf outlines. In the center, the words "THANK YOU" are written in a large, dark blue, serif font. The text is slightly shadowed, giving it a three-dimensional appearance as if it's floating above the background.

THANK
YOU